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GROUP SPECIFICATION

Experiential Networked Intelligence (ENI); ENI Requirements

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Foreword

This Group Specification (GS) has been produced by ETSI Industry Specification Group (ISG) Experiential Networked Intelligence (ENI).

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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1 Scope

The purpose of the present document is to specify the requirements of how intelligence is applied to the network and applications in different scenarios to improve experience of service provision and network operation. Also, how intelligence enables dynamic autonomous behaviour and adaptive policy driven operation in a changing context. The ENI requirements are based on the ENI use case document and identified requirements from other Standard Developing Organizations (SDOs). These requirements will form the base for the architecture design work.

In version 4, the present document will add the requirements abstracted from new use cases and PoCs, as well as from the deliverables of other version 4 ENI work.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found in the ETSI docbox.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI GS ENI 001: "Experiential Networked Intelligence (ENI); ENI use cases".

[2] ETSI GS ENI 005: "Experiential Networked Intelligence (ENI); ENI System Architecture".

[3] ETSI TS 102 165-1 (V5.2.5): "CYBER; Methods and protocols; Part 1: Method and pro forma for Threat, Vulnerability, Risk Analysis (TVRA)".

[4] ETSI GS NFV 006 (V4.4.1): "Network Functions Virtualisation (NFV) Release 4; Management and Orchestration; Architectural Framework Specification".

[5] ETSI TS 101 158: "Telecommunications security; Lawful Interception (LI); Requirements for network functions".

[6] ETSI GS ENI 019 (V4.1.1): "Experiential Networked Intelligence (ENI); Representing, Inferring, and Proving Knowledge in ENI Models.

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] ETSI GR ENI 004: "Experiential Networked Intelligence (ENI); ENI terminology".

[i.2] Service Operations Specification MEF 55: "Lifecycle Service Orchestration (LSO): Reference Architecture and Framework".

[i.3] Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation).

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in ETSI GR ENI 004 [i.1] apply.

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

AI	Artificial Intelligence
API	Application Programming Interface
BP	Back Propagation
BSS	Business Support System
CAP	Context Aware related Policy
CAPEX	CAPital Expenditure
CPU	Central Processing Unit
DC	Data Collection

NOTE: Used in the context of servers.

DCA	Data Collection and Analysis
DSL	Domain-Specific Language
EMS	Element Management System
ENI	Experiential Networked Intelligence
GDPR	General Data Protection Regulation
GPM	General Policy Management
GPU	Graphic Processing Unit
IoT	Internet of Things
IP	Internet Protocol
IT	Information Technology
KPI	Key Performance Indicator
LI	Lawful Interception
LLC	Last Level Cache
LSO	Lifecycle Service Orchestration
LSO RA	Lifecycle Service Orchestration - Reference Architecture
MANO	MANagement and Orchestration
MEC	Multi-access Edge Computing
MEF	Metro Ethernet Forum
MOP	Mode Of Operations
NFV	Network Functions Virtualisation
NM	Network Model
NPD	Network Planning & Deployment
OPEX	OPerational EXpenditure
OR	Operational Requirements
OSS	Operational Support System
PINet	Polymorphic Network
PR	Performance Requirements
RA	Reference Architecture

RR	Resilience and Reliability
SDN	Software Defined Networking
SDO	Standard Developing Organizations
SLA	Service Level Agreement
SOM	Service Orchestration and Management
SP	Security and Privacy
SP.2A	Security and Privacy 2A
SP.2B	Security and Privacy 2B
SVM	Support Vector Machine
TCO	Total Cost of Ownership
TVRA	Threat, Vulnerability and Risk Analysis
VNF	Virtualised Network Function

4 Introduction

4.1 Categorization of the requirements

The present document structure addresses the requirements in the following areas:

1) Service and network requirements:
   - General requirements
   - Service orchestration and management
   - Network planning and deployment
   - Network optimization
   - Resilience and reliability
   - Security and privacy
2) Functional requirements:
   - Data collection and analysis
   - Policy management
   - Data learning
   - Interworking with other systems
   - Mode of operations
   - Model training and iterative optimization
   - Mode of deployments
   - API requirements
3) Non-functional requirements:
   - Performance requirements
   - Operational requirements
   - Regulatory requirements

5 Service and network requirements

5.1 Overview

The requirements in this clause are addressed from service and network point of view and are derived directly from the related use cases.

5.2 General requirements

This clause captures the requirements that are general and independent from other requirements categorized in other specific clauses.

[GR.1] The ENI System shall use history data, context, and decisions taken to learn, process and provide responses to events, whether generated from devices or from management systems.

[GR.2] The ENI System shall use context information as part of the computations that result in recommendations, advisement, predictions, and decisions that are used to assist other network systems, e.g. orchestration and management systems.

NOTE: As an example, MANO (from ETSI GS NFV 006 [4]) or the LSO RA (from MEF [i.2]) are different types of orchestration and management systems.

[GR.3] The ENI architecture shall be flexible enough to support extensibility.

5.3 Service orchestration and management

This clause captures requirements related to the ENI System service provisioning, e.g. how to compile the service intent and orchestrate the service atoms and workflows, as well as automatic service on boarding.

[SOM.1] The ENI System shall invoke policies based on models that describe and/or define traffic behaviour, such as SLAs (e.g. past or current telemetry).

[SOM.2] The ENI System shall support the closed loop control model when different orchestration and management systems are used.

NOTE 1: As an example, MANO (from ETSI GS NFV 006 [4]) and LSO RA (from MEF [i.2]) are different types of orchestration and management systems.

[SOM.3] The ENI System should not directly manage, control or orchestrate physical or virtual entities, either at the infrastructure level or service level.

NOTE 2: ENI System may interact with the Orchestration system, EMS or OSS/BSS to influence the state of the resources or services.

5.4 Network planning and deployment

This clause captures requirements related to network planning and deployment, e.g. how to allocate network resources to VNFs, or automatic VNF on boarding.

NOTE 1: The network resources that can be managed are not limited to the requirements addressed in this clause.

[NPD.1] The ENI System shall recommend allocation or retrieval of network resources, e.g. virtual machines, bandwidth, IPv4 addresses and IPv6 prefixes to end users or service flows, in an intelligent way to improve the efficiency of resource utilization. This ENI System function shall be implemented in a centralized and/or distributed manner, according to what is defined in ETSI GR ENI 004 [i.1] and according to ETSI GS ENI 005 [2].

[NPD.2] The ENI System shall assist the network equipment or network function to use the resource pools that are used for resource allocation (e.g. virtual machines, network bandwidth, IP addresses, CPU cores, CPU frequencies, memory, memory bandwidth), in an intelligent way in order to improve the efficiency of resource utilization and user experiences.

[NPD.3] The ENI System should dynamically and intelligently compute and recommend the required network resources, including both IPv4 and IPv6 resources as well as other resources.

[NPD.4] The ENI System shall compute the network resources required to dynamically and intelligently deploy a given network service efficiently.

[NPD.5] IT resources to enable network services shall be managed within the ENI System.

NOTE 2: Similar capabilities within the data centre are outside the network scope of this phase of ENI.

[NPD.6] The ENI System shall be capable of understanding the context that a set of devices is operating within.

[NPD.7] The ENI System shall be capable of performing the proper planning and deployment of resources to ensure that applicable deployed policies are not violated.

[NPD.8] The ENI System shall identify different types of rollouts for different types of resources that lead to the upgrade of virtualised software-based resources.

[NPD.9] The ENI System shall, in an efficient and dynamic manner, combine network slices, slice/service prioritization and resource allocation concepts, e.g. in order to resolve resource allocation conflicts between competing network slices deployed on top of a shared infrastructure.

[NPD.10] The ENI System shall support cross-domain and global resource management, e.g. cross-domain node content resource aggregation and integration, as well as cross-domain permissions management of unified user identity.

[NPD.11] The ENI System shall dynamically map terrestrial and satellite network slices using AI-driven prediction (e.g. Graph Convolutional Networks (GCNs) and Gated Recurrent Units (GRUs) GCN-GRU models) to forecast resource demands and ensure end-to-end service continuity across heterogeneous domains.

[NPD.12] The ENI System shall deploy programmable slicing gateways to translate data-plane identifiers (e.g. VLAN/IP) between terrestrial and satellite networks, enforcing cross-domain traffic policies based on ENI directives.

[NPD.13] The ENI System shall orchestrate control-plane coordination between terrestrial (e.g. The 5G Core 5GC) and satellite (e.g. Security Network Operations Center SNOCC) domains to resolve slicing protocol mismatches and establish unified end-to-end sessions.

[NPD.14] The ENI System shall dynamically resolve resource allocation conflicts between competing slices in shared space-ground infrastructures by combining service prioritization, predictive scaling, and cross-domain arbitration.

[NPD.15] The ENI System shall adapt slice mappings during UE session initiation based on service type, QoS class, and predicted resource availability across terrestrial and satellite segments.

5.5 Network optimization

This clause captures requirements related to network optimization, e.g. how to adjust the network configurations to improve its efficiency and performance, as well as the user experience of the service.

[NO.1] The ENI System shall collect and process the necessary data according to specific algorithms in order to achieve network optimization.

NOTE 1: Data collection and processing algorithms for systems will be specified in the functional architecture.

[NO.2] The ENI System shall meet or exceed all performance requirements when improving the target performance.

[NO.3] The ENI System shall support central optimization, local optimization and distributed joint optimization, according to what is defined in ETSI GR ENI 004 [i.1].

[NO.4] The ENI System shall support an adaptive optimization process where changes in the environment are reflected in the results of the optimization.

[NO.5] The ENI System shall use prioritization and other scheduling and traffic shaping techniques to prevent SLA violations related with priority services.

[NO.6] The ENI System shall use AI (e.g. Machine Learning) to identify traffic type and support traffic handling and QoS assurance for specific type of traffic.

[NO.7] The ENI System shall support traffic type identification in different granularity levels, including application types, action types (e.g. sending pictures, voice calls, etc.).

[NO.8] The ENI System shall support dynamic policy adjustment, i.e. to select or re-select and enforce a policy dynamically, for a specific flow based on traffic identification results.

NOTE 2: Applying policies to specific flows may lead to hundreds of thousands of policies.

[NO.9] The ENI System shall dynamically optimize beam patterns and spectrum allocation in multi-beam satellite systems using AI-driven traffic prediction and Deep Reinforcement Learning (DRL).

[NO.10] The ENI System shall seamlessly steer traffic between satellite and terrestrial networks during link failures or congestion events, ensuring handover latency $\leq$ 20 ms and uninterrupted service continuity.

[NO.11] The ENI System shall prioritize critical services (e.g. emergency communications) through context-aware resource allocation, dynamically adjusting beam bandwidth and coding schemes (e.g. LDPC) based on real-time SLA thresholds.

[NO.12] The ENI System shall predict and resolve coverage gaps in remote/aerial/maritime regions by correlating UE density spikes, historical traffic patterns, and satellite orbital dynamics using GCN-GRU spatiotemporal models.

[NO.13] The ENI System shall retrain DRL models iteratively using real-world KPI feedback (throughput, latency, packet loss) to maintain optimization accuracy amid changing network topologies and traffic profiles.

5.6 Resilience and reliability

This clause captures requirements related to resilience and reliability of the network, including fault diagnosis and prediction, high availability and back up, conflict detection, and rolling back to previous policies and status.

[RR.1] The ENI System shall intelligently recommend allocation or retrieval of IP addresses without causing route oscillation.

[RR.2] The ENI System shall intelligently recommend allocation or retrieval of IP addresses without causing any interruption in the offered services.

[RR.3] The ENI System shall support root cause analysis to diagnose existent faults and potential faults caused by new cases, according to what is defined in ETSI GS ENI 005 [2].

[RR.4] The ENI System shall support the use of one or more AI algorithms to perform network service fault prediction.

[RR.5] The ENI System shall learn and predict the pattern of resource requirements of services.

[RR.6] When optimization of energy consumption is required, which implies a switch of servers, the ENI System shall trigger the reallocation of services to appropriate resources in another server.

[RR.7] The ENI System shall wake up an appropriate number of servers in time to meet the growing resource needs required by services, after learning and predicting the pattern of resource requirement of those services.

[RR.8] The ENI System shall provide the operators with the ability to define services that are critical or prioritized.

[RR.9] The ENI System shall allow the on-going services in a server to be moved from this server to another without interruption, e.g. during reallocation for energy saving purposes.

[RR.10] The ENI System shall not interrupt the on-going services on the target servers, e.g. when reallocation of services from other servers takes place for energy saving purposes.

[RR.11] Energy saving need not be the only criterion for moving a service.

[RR.12] The ENI System shall calculate and propose proper backup actions to the operators in order to prevent or to mitigate a service degradation or disruption when a planned operation occurs.

[RR.13] The ENI System shall support the use of one or more intelligent methods to perform network anomaly (fault, error and unusual behaviour) prediction and prevention.

[RR.14] The ENI System shall be aware of the impact of adjustment on services and guarantee seamless adjustment of network slice and high valued services.

5.7 Security and privacy

This clause captures requirements related to security and privacy issues (e.g. it is recommended that data collection shall be captured in a secure way and not add more security risks). In addition, it is recommended that the collected data shall be accessible by authorized accounts, and that the privacy of both subscribers and operators are protected.

The requirements indicated in the present document have been derived from application of the ETSI TVRA method defined in ETSI TS 102 165-1 [3], the details of the analysis leading to the requirements have been examined with respect to the use cases defined in ETSI GS ENI 001 [1] and with respect to the terminology defined in ETSI GR ENI 004 [i.1].

[SP.1] The ENI System shall use AI (e.g. Machine Learning) to detect abnormal traffic patterns that can lead to service disruptions or security threats as well as to carry out the identification of abnormally operating devices.

[SP.2] The ENI System shall provide means to detect a corrupted device.

[SP.2A] The ENI System shall provide means to identify a corrupted device.

[SP.2B] The ENI System shall provide means to isolate and remove a corrupted device from the system.

[SP.3] The ENI System shall provide means to indicate to authorized parties the occurrence of potential and confirmed security threats by using appropriate mechanisms, (e.g. via dedicated interfaces).

[SP.4] The ENI System shall provide means to invoke policies to isolate threats.

[SP.5] The ENI System shall be designed in such a way that it complies to the provisions of the GDPR [i.3], when processing of data (traffic or signalling).

[SP.6] The ENI System shall allow entities that are involved in services that are subject to LI processing to be designed in such a manner that they comply with the general provisions of LI as defined in ETSI TS 101 158 [5].

[SP.7] Processing for security functions (e.g. access control, user authentication, privacy protection mechanism) shall always be enabled.

[SP.8] The addition of any processing in the ENI System to comply with provisions arising from LI compliance shall not be visible to an external observer.

NOTE: The consequence of the above is that an external observer should not assert that an LI operation is taking place by observation of the processing load of the ENI System.

6 Functional requirements

6.1 Overview

The requirements in this clause are addressed from the architecture point of view.

6.2 Data collection and analysis

This clause captures requirements related to how data is collected and analyzed by the ENI System.

[DCA.1] The ENI System shall gather network status data (e.g. related to connection or routing protocols in use) as well as network operational, administrative, and state information (e.g. network configuration, network topology of data link, network node working status, resource storage information).

[DCA.2] The ENI System shall store the data either as raw data or aggregated data for further analysis, according to what is defined in ETSI GR ENI 004 [i.1].

[DCA.3] The ENI System shall provide data analysis functionalities, which make use of collected data in order to produce intermediate information that will support further analysis.

NOTE 1: Examples of information related to this requirement include network context information (such as time of the day, device/link state, and location of users).

[DCA.4] The ENI System shall collect and analyse the necessary data in order to determine traffic patterns.

NOTE 2: This requirement can be governed by national and international regulations on Data Protection and Privacy.

[DCA.5] The ENI System shall collect information from the infrastructure, e.g. massive-MIMO array antenna information such as geometrical configuration of overall array, relative power pattern of the individual element and relative displacement of the elements, RF parameters or code book with corresponding antenna power pattern.

[DCA.6] The ENI System shall collect, store and pre-process the history data, and extract features from it, to be used for e.g. further analysis, learning process, real time prediction of future business requirement and network status, etc.

[DCA.7] The ENI System shall collect and store required run-time data in order to e.g. determine the policy.

[DCA.8] The ENI System shall provide data collection methods for network KPI status data (e.g. packet loss rate, latency, throughput) in different granularities (e.g. physical interface, logical interface, flow).

[DCA.9] The ENI System shall perform analysis on the combination of data, collected from various relevant infrastructure elements, to generate an overall view of network status.

[DCA.10] The ENI System shall be capable of requesting or retrieving certain kind of data from the Assisted System.

[DCA.11] The ENI System shall be capable of extracting configuration information from the collected data to generate policies intelligently (e.g. avoiding adjust DC servers that are unavailable or inaccessible).

[DCA.12] The ENI System shall automatically assign labels to new samples.

[DCA.13] The ENI System shall be able to collect service data e.g. service performance log data, service workload data and resource configuration of services for further analysis and policy making.

[DCA.14] The ENI System shall be able to collect fine grained telemetry data to provide enhanced visibility into how the infrastructure is operating and how the application is using its resources, e.g. how shared resources such as Last Level Cache (LLC) and memory bandwidth are used.

[DCA.15] The ENI System shall be able to support the pre-processing, integration, analysis and storage of multi-source and heterogeneous data (e.g. compatibility of data from different Network Model (NM) e.g. IP based NM, Identity based NM, Content based NM and Location based NM in Polymorphic Network (PINet)).

NOTE 3: Heterogeneous data pre-processing may include heterogeneous data discrimination, sampling and cleaning. Heterogeneous data integration may include heterogeneous data label fusion, heterogeneous data fusion analysis, and heterogeneous data feature extraction. Clustering analysis, association analysis and classification prediction algorithms can realize the fusion analysis of structured data and unstructured heterogeneous data from multiple sources.

6.3 Policy management

6.3.1 General policy management requirements

This clause captures requirements related to how policies are managed by the ENI System.

The ENI System enforces policies. Policies should be used to manage the behaviour and operation of the Assisted System. The actions shall be either a set of commands or recommendations, depending on the mode that the ENI System is operating in. The ENI System shall use any combination of imperative, declarative and intent policies to form the actions, see ETSI GS ENI 005 [2].

[GPM.1] The content of the recommendations and/or commands shall be changed through the execution of policies that affect the ENI System.

[GPM.2] Pre-defined policies that are triggered by one or more events shall be executed by the ENI System while complying with any metadata or other policies that govern their operation.

[GPM.3] When an event indicates that a policy violation is detected, the ENI System should be aware and then take appropriate remediation, or stop execution and ask the policy author what to do.

[GPM.4] The ENI System shall be capable of performing policy conflict resolution in order to ensure that the overall system behaviour complies with all policies.

[GPM.5] The ENI System shall use all relevant external information, including but not limited to metadata, business rules, context, and regulatory requirements, to achieve the goals of the Assisted System.

[GPM.6] The ENI System shall detect changing conditions in the network and adapt the behaviour and operation of the Assisted System using policies.

[GPM.7] The ENI System shall send policies to external entities by using dedicated External Reference Points.

[GPM.8] The ENI System shall be capable of translating a natural language intent into a set of ENI Policies.

6.3.2 Context aware related policy requirements

This clause is based on policy related functional requirements defined in ETSI GS ENI 005 [2].

[CAP.1] In the ENI System, a single information model shall be used to represent the structure and semantics of policies.

[CAP.2] Multiple data models shall be derived from the (single) information model.

[CAP.3] The Policy information model shall be integrated into the system's information model. This facilitates relating policies to Services, Resources, Functional Blocks, and other managed entities of the system.

[CAP.4] The information model shall be used in constructing APIs and DSLs.

[CAP.5] In the ENI System, different types of policies shall be supported in the same system.

[CAP.6] Policy within a domain shall act as a mechanism to define and manage the behaviour of entities contained within that domain. This includes entities in a sub-domain contained in a higher-level domain.

[CAP.7] Policy defined in peer domains shall act as a mechanism to negotiate mutually acceptable behaviour between the domains.

[CAP.8] Situations shall be used to select active policies (and deactivate policies that are no longer appropriate). This enables the ENI System to adjust its goals and behaviours accordingly to changes in context.

[CAP.9] The ENI System, based on contextual changes, shall construct new policies to address problems found and/or improve system behaviour by using operator intervention if no suitable policies currently exist.

NOTE 1: According to what is defined in ETSI GR ENI 004 [i.1], a policy may be cloned and modified by an applicable functional block, such as policy engine (as opposed to the policy itself being self-modified). The changes allowed by this process will be limited according to parameters defined in the ENI architecture. Self-modifying policies are still in the early research stage; in this stage of ENI, self-modifying policies are not specified.

NOTE 2: CAP.9 requirement is to be supported in a future release.

[CAP.10] Policy shall be augmented by metadata.

[CAP.11] Policy metadata should be used to describe and/or prescribe functionality.

[CAP.12] Policy shall be realized by using centralized and/or distributed architectures, according to what is defined in ETSI GS ENI 005 [2].

[CAP.13] The (single) information model, and each data model, should use a set of software design patterns to provide extensibility and consistency.

[CAP.14] The derivation of a data model should be done by defining formal transformations from the (single) information model to each different data model. This facilitates the development of associated software tooling.

[CAP.15] Different entities shall be represented by different roles being mapped to different levels in the Policy Continuum.

[CAP.16] Different entities have different concepts and terminologies for their policies. Therefore, the notion of a Policy Continuum, where each continuum corresponds to a given set of entities, should be realized.

[CAP.17] Formal mathematical transformations between each level of the Policy Continuum and its entities should be defined, which facilitates the development of APIs and DSLs.

[CAP.18]: The ENI System should be prepared to modify the role of outside systems.

NOTE 3: Considering that entities operate in a changeable context throughout the lifecycle of the system. Therefore, the context of an entity may be used to map the activities of the entity to a particular level in the Policy Continuum. In this context, CAP.18 applies.

NOTE 4: An end-user is always an end-user, and is defined by that particular MCMPartyRole. If that same end-user wants to be a Developer, then that end-user will assume a new MCMPartyRole. (This behaviour is fundamental to the role-object pattern, and is defined in ETSI GS ENI 005 [2]).

[CAP.19] The ENI System should use historical data about contextual changes along with other collected data to perform analysis and generate policies.

6.4 Data learning

This clause captures requirements related to data learning in ENI System.

[DL.1] The ENI System shall provide basic data learning functionalities, such as classification, regression, and clustering.

NOTE: The ENI System may contain common algorithms to support these functionalities, such as SVM for classification, BP for regression and K-Means for clustering.

[DL.2] The ENI System shall provide feature analysis functionalities, which can rank all features of the data in the order of their importance and provide suggestions to operators for their further feature weighting.

[DL.3] The ENI System shall provide big data mining and analysis functionalities, which can find the hidden patterns, correlations and knowledge and reveal the relationship between correlation rules and events through mining and analysing large amounts of data.

[DL.4] The ENI System shall provide data annotation functionality, which can classify and organize the data to be standardized, and provide data coding to meet data analysis requirements.

6.5 Interworking with other systems

This clause captures requirements related to how the ENI System interworks with other systems.

[IWOS.1] The ENI System shall interwork with other systems, e.g. NFV, MEC, SDN, MEF LSO, DC Dynamic Environment System, OSS, and others that shall show up in a 5G-Network context, by reusing existing functions and interfaces as much as possible.

[IWOS.2] The failure of ENI System shall not interrupt the services provided by other systems.

6.6 Mode of operations

This clause captures requirements on the mode of operations of the ENI System.

[MOP.1] The ENI System shall support a recommendation mode.

[MOP.2] The ENI System shall support a management mode of operation.

[MOP.3] The ENI System shall support a mixed mode of operation for different sets of decision classes (e.g. some types of decisions are made using one mode of operation, and other sets of decisions are made using the other mode of operation).

[MOP.4] The ENI System shall discover (and/or be told) the capabilities of the Assisted System in supporting the desired mode of operation.

[MOP.5] The ENI System shall support and adapt to external inputs for each mode of operation.

[MOP5.A] The ENI System shall support and adapt to changes in the context and/or situation of the Assisted System.

[MOP5.B] The ENI System shall support external input of regulatory policies and operator goals.

[MOP.6] The ENI System shall use the above two factors in [MOP5] to select its mode of operation.

[MOP.7] The ENI System shall ask permission from the Operator or Designated Entity to change modes of operation using an agreed External Reference Point.

[MOP.8] The Assisted System, or its Designated Entity shall ask the ENI System to change modes of operation using an agreed External Reference Point.

[MOP.9] The ENI System shall confirm through the agreed External Reference Point to the Operator or Designated Entity of the Assisted System when it has successfully switched modes of operation.

[MOP.10] The ENI System shall suggest that a particular mode of operation is used when a class of decision is reached that is not specified by the Assisted System.

[MOP.11] The Assisted System and/or its Designated Entity need not accept the recommendations offered by the ENI System when in recommendation mode. This includes those decisions that apply to recommendations when the Assisted System is in a mixed mode of operation.

[MOP.12] The Assisted System and/or its Designated Entity need not accept the recommendations offered by the ENI System when in management mode. This includes those decisions that apply to commands when the Assisted System is in a mixed mode of operation.

[MOP.13] Decisions and commands in management mode shall be subject to the approval of the Assisted System (or its Designated Entity).

[MOP.14] The Assisted System (or its Designated Entity) shall tell the ENI System that it approves all commands sent to it when it is in management mode.

[MOP.15] The Assisted System shall revoke the above setting at any time, in which case [MOP.13] then applies (once acknowledged by the ENI System).

[MOP.16] If the Assisted System (or its Designated Entity) rejects a command set to it by the ENI System when it is in recommendation mode, it shall send a notification to the ENI System.

[MOP.17] If the Assisted System (or its Designated Entity) rejects a command set to it by the ENI System when it is in management mode, it shall send a notification to the ENI System.

6.7 Model training and iterative optimization

[MTIO.1] The ENI System shall be able to support the selected algorithms according to the types of problems, e.g. classification, regression, etc., and support the initialization of model parameters and structure based on data features and task objectives.

[MTIO.2] The ENI System shall support the ability to adjust a model's hyperparameters or choose a new model.

[MTIO.3] The ENI System shall re-train the model using various learning techniques and algorithms through continuously extending the existing model's knowledge according to input data.

NOTE: Learning techniques may include Online Learning and Offline Learning techniques, including Incremental Learning, Transfer Learning with their associated algorithms, etc.

[MTIO.4] The ENI System shall solve related problems based on knowledge gained while solving one or more previous problems.

6.8 Model inference

[MI.1] The ENI System shall support inference with the pre-trained models using the data fed into the system.

[MI.2] The ENI System shall support inference with the pre-trained models on different types of AI hardware, including CPUs, GPUs, and AI acceleration cards, etc.

[MI.3] The ENI System shall support the use of assistant migration tools to migrate models trained on GPUs to CPUs for inference.

6.9 Mode of deployments

[MOD.1] The ENI System shall support centralized deployment and distributed deployment.

[MOD.2] Information exchange shall be supported between distributed ENI Systems located in the same network domain.

[MOD.3] Information exchange shall be supported between distributed ENI Systems located in different network domains.

[MOD.4] Information exchange shall be supported between distributed ENI Systems and one or more centralized ENI Systems.

6.10 API requirements

[API.1] The ENI System shall support API based programming mode.

NOTE 1: If [API.1] is used then the following corollaries follow:

[API.1A] The ENI System shall provide a set of common APIs in driver level to support AI development across a range of data parallel accelerators.

[API.1B] The ENI System shall provide a set of APIs to support AI development across different AI frameworks.

[API.1C] The ENI System shall reuse the APIs provided by an existing external system as external ENI APIs, if these APIs can meet the requirement of interworking between ENI System and the external system.

[API.2] The ENI System should support direct programming mode if the API based programming mode cannot meet the required needs.

NOTE 2: [API.1] and [API.2] can be used in combination.

NOTE 3: API based programming mode is the way that programmers produce code to achieve particular function by calling existing APIs. Direct programming mode is the way that the programmers use programming language to produce code to achieve particular function without calling existing APIs.

[API.3] The ENI System shall be able to communicate with external systems by a single set of ENI APIs directly or via an optional API Broker.

6.11 AI Agents Requirements

[AIAR 1-1] Subject to operator policy and regulatory requirements, the mobile network shall be used to contact AI-Core, which then provides a mechanism to uniquely identify an AI Agent that acts on behalf of the user.

[AIAR 1-2] The mobile network shall use a combination of authentication, opaque execution, and policy enforcement to ensure that all AI Agents preserve the privacy of the owner (e.g. user, the 3rd party) when they exchange information with AI Agents.

NOTE 1: Opaque execution may be a cornerstone of modern multi-agent communication protocols. It means that when one agent asks another to perform a task, the second agent may be able to complete the task without revealing its internal methods, data, or reasoning processes. The collaboration may happen through well-defined, standardized interfaces, but the "inner workings" of each agent remain private.

NOTE 2: When sensitive information needs to be exchanged, robust agentic systems may have guardrails for sensitive data handling and isolation. This means there may be explicit policies that govern how sensitive data is used and shared between agents. Privacy preservation may be the set of rules followed during the conversation, ensuring that sensitive personal or business information is not revealed unnecessarily.

[AIAR 1-3] The architecture includes a standardized "integration fabric" that shall mediate all multi-agent communication. This fabric shall support multiple interaction patterns, such as request-response for direct commands and publish-subscribe for asynchronous event streaming, enabling decoupled and scalable collaboration.

[AIAR 1-4] The system shall provide a service registry or Knowledge Graph (KG) that allows agents to dynamically discover and consume network-provided services. These services augment agent capabilities shall be realized by offering specialized data (e.g. high-precision location) or AI models (e.g. congestion prediction) as a service.

[AIAR 1-5] A high-level orchestrator agent shall be responsible for end-to-end task management. It uses formal planning techniques, such as Hierarchical Task Networks (HTNs), to decompose high-level intents into sub-tasks, which are then delegated to specialized agents based on their advertised capabilities.

[AIAR 2-1] The system shall provide a standardized interface for authorized users to initiate service requests by submitting high-level, declarative intents. This interface shall abstract the underlying complexity, allowing users to define their desired outcomes without specifying the implementation details.

[AIAR 2-2] Upon receiving a validated intent, the system's AI Orchestration function shall provide the requested service on-demand. This shall be achieved by autonomously planning and decomposing the intent into a workflow of sub-tasks, which are then delegated to the appropriate specialized agents for execution.

[AIAR 3-1] The Intent Management Entity of the AI-Core shall expose a standardized, secure interface over the Integration Fabric that is accessible to authenticated user agents on mobile.

[AIAR 3-2] All interactions between the AI Phone agent and the AI-Core, transmitted over the mobile network, shall be encrypted and authenticated to prevent eavesdropping or man-in-the-middle attacks. This is a foundational security posture for any trusted system.

[AIAR 3-3] The AI-Core shall maintain a discoverable registry of available services and their capabilities, allowing the AI Phone agent to query which network-provided functions (e.g. data fusion, design generation, QoS management) it can invoke.

[AIAR 4-1] The system's AI Orchestrator shall be capable of dynamically composing and deploying logical networks (i.e. network slices) on-demand. This process integrates network capabilities (e.g. QoS guarantees) with application-layer functions (e.g. AI-based video analytics) by decomposing a user's intent and chaining the required services from both network and third-party domains.

NOTE 3: The generated networks are not pre-designed and are generated on-the-fly by the multi-agent tailored to the requests of users. The logic networks do not only include the network capabilities/functions but also include the application functions according to the request of users.

[AIAR 4-2] The AI Orchestrator shall autonomously manage the entire lifecycle of the requested service. This includes using Hierarchical Task Network-based planning to design the service chain, delegating deployment tasks to specialized agents, monitoring execution, and ensuring the complete reclamation of all resources upon service termination.

[AIAR 4-3] The system shall implement a closed-loop assurance mechanism. The AI Orchestrator needs to monitor cross-domain telemetry in real-time to verify that the operational state of the logical network continuously complies with the performance and QoS constraints defined in the original intent.

[AIAR 4-4] The AI Orchestrator shall be capable of self-optimization. When the assurance loop detects a performance degradation or potential SLA violation, the orchestrator needs to autonomously trigger corrective actions, such as re-allocating resources or adjusting network configurations, to restore compliance with the user's intent.

[AIAR 5-1] The system's AI Orchestrator shall be capable of dynamically composing and deploying logical networks that co-provision network services (e.g. connectivity with specific QoS) and application-layer functions (e.g. data processing) as a single, integrated on-demand service.

[AIAR 5-2] The system shall support the autonomous, end-to-end lifecycle management of the composed service. This includes the initial design and deployment based on intent, runtime monitoring and modification, and the complete recycling of all network and compute resources upon service termination, all managed by the AI Orchestrator.

[AIAR 5-3] The system shall provide closed-loop assurance by continuously monitoring the performance of the deployed network and application functions. The AI Orchestrator is required to autonomously detect and remediate any deviations from the user's intent to ensure correct execution and SLA compliance.

[AIAR 5-4] In the case of Non-Public Networks (NPN), where the network service is not provided by a public Mobile Network Operator, the mobile network shall be usable in a nomadic node manner. This results in the fact that the cellular infrastructure is highly mobile and can be used at multiple various construction sites.

[AIAR 6-1] The system shall include an Intent Translation Pipeline that leverages Natural Language Processing to parse high-level, unstructured user intents into a formal, machine-readable specification for the orchestration layer.

[AIAR 6-2] The architecture shall use an AI Orchestrator to manage the end-to-end service lifecycle and an Integration Fabric to facilitate secure, standardized communication between internal network agents and external third-party agents.

[AIAR 6-3] The system shall support closed-loop assurance and optimization, where the AI Orchestrator continuously monitors service performance against the user's intent and autonomously triggers corrective actions. This requires a Service Registry where agents can discover each other and their capabilities.

[AIAR 6-4] The system shall expose its network and AI capabilities (e.g. QoS management, predictive analytics) as discoverable, well-defined services on the Integration Fabric, allowing authorized external agents to invoke them via standardized APIs.

[AIAR 7-1] The AI-Core shall provide a Service Registry that allows authorized agents representing energy sites to register and publish their operational capabilities to the network's KG in a standardized format.

[AIAR 7-2] The system shall expose a secure Intent Management Interface that allows authorized operators to initiate energy optimization services by submitting high-level, declarative intents.

[AIAR 7-3] The architecture shall support a hierarchical agent model. A central AI Orchestrator is required to perform global, system-wide optimization and planning, while delegating real-time data processing and task execution to lightweight agents deployed at the network edge and on-site.

[AIAR 7-4] The AI Orchestrator shall be capable of translating high-level intent into an executable workflow. This includes analysing contextual data from the KG, generating an optimal coordination strategy, delegating tasks to subordinate agents, and issuing derived intents to network domain managers to configure the required QoS.

[AIAR 7-5] The AI Orchestrator shall implement a closed-loop assurance function, continuously monitoring execution outcomes and using this feedback to dynamically learn and adapt its energy distribution strategies to persistently meet the user's intent.

[AIAR 8-1] The system shall expose a single, unified Intent Management Interface as the northbound entry point for all management tasks. This interface needs to be supported by an Intent Translation Pipeline capable of parsing natural language requests to abstract away the complexity of the underlying infrastructure.

[AIAR 8-2] The system's AI Orchestrator shall be capable of fulfilling user objectives by autonomously translating a validated intent into an executable workflow, decomposing it into sub-tasks, and delegating those tasks to the appropriate specialized agents.

[AIAR 8-3] The architecture shall support hierarchical multi-agent collaboration for closed-loop operations. A central AI Orchestrator needs to manage the overall workflow, delegating specific functions like exception detection, root cause analysis, and remediation to distinct, specialized worker agents that collaborate to achieve the system's goal.

[AIAR 9-1] The system's Integration Fabric shall support both hierarchical communication patterns for centralized "peacetime" orchestration and robust, decentralized Peer-to-Peer (P2P) protocols to enable agent collaboration when the network is fragmented and central control is lost.

[AIAR 9-2] The system shall include a specialized Analytics Agent capable of performing predictive analysis on network telemetry and external data (e.g. weather patterns) to forecast the impact of potential disasters and proactively trigger preparatory actions.

[AIAR 9-3] AI Agents deployed at the network edge shall be empowered with the autonomous capability to reconfigure local network elements and instantiate essential network functions. This enables them to restore connectivity and services within an isolated segment without real-time instruction from a central orchestrator.

[AIAR 10-1] The architecture shall support Monitoring Agents capable of collecting granular, real-time telemetry from multi-RAT RAN nodes and other network domains, streaming this data to a central Knowledge Graph via the Integration Fabric.

[AIAR 10-2] All agents and network nodes involved in the TSN control plane shall be synchronized to a unified, high-precision network clock. This is a foundational requirement for calculating and enforcing the deterministic schedules essential for time-sensitive applications.

[AIAR 10-3] Specialized agents, under the direction of the AI Orchestrator, shall have the capability to configure low-level control parameters on network nodes to manage TSN schedules, coordinate resources, and enforce QoS policies.

[AIAR 10-4] The system's Integration Fabric shall provide secure and reliable communication channels that enable specialized agents across different domains (e.g. RAN, transport) to share state information and collaboratively plan end-to-end service configurations.

[AIAR 10-5] The AI Orchestrator shall include an Intent Reconciliation Function capable of detecting and resolving conflicts between multiple TSN service requests that compete for the same limited network resources, ensuring deterministic service for all accepted intents.

[AIAR 11-1] Core Network Functions (e.g. AMF, SMF, PCF) shall expose standardized, real-time telemetry interfaces to allow authorized Analytics Agents (such as the NWDAF) to monitor control-plane signalling metrics.

[AIAR 11-2] Core Network Functions shall provide secure, programmable APIs that allow an authorized Orchestrator Agent to dynamically adjust signalling policies and parameters (e.g. timers, registration areas) in real time.

[AIAR 11-3] The system's Intent Management Interface shall accept high-level operational intents (e.g. "minimize signalling load for IoT devices"). The AI Orchestrator is responsible for directing the translation of these intents into specific, actionable optimization strategies.

[AIAR 11-4] The system shall support a workflow where an Analytics Agent infers signalling inefficiencies from network data, and an Orchestrator Agent autonomously enacts the corresponding optimizations by issuing configuration and policy commands to the relevant Core Network Functions.

[AIAR 11-5] The system shall implement a closed-loop assurance mechanism that continuously measures the impact of signalling optimizations and uses the performance outcomes as feedback to refine the underlying AI models for future decisions.

[AIAR 12-1] The system shall provide authorized Analytics Agents with secure access to a comprehensive, cross-domain dataset. This includes real-time telemetry from network functions as well as relevant customer and service data from the operator's OSS/BSS and IT domains, ideally unified within a query able KG.

[AIAR 12-2] The AI Orchestrator shall be empowered to enact decisions by issuing dynamic configuration and policy adjustments to Core Network Functions (e.g. PCF, SMF). It also needs to be able to manage the lifecycle of the underlying virtualized resources, such as scaling containerized network function instances, to meet service demands.

[AIAR 12-3] The system shall implement a closed-loop assurance mechanism. The AI Orchestrator and Analytics Agent need to continuously verify that optimization actions have successfully improved the user experience and use these performance outcomes as feedback to refine their decision-making models.

6.12 Information Model Requirements

The following clauses define requirements for the ENI information model.

[IM01] The ENI Information Model shall use single inheritance.

[IM02] The ENI Information Model shall not use multiple inheritance.

[IM03] The ENI Information Model shall not use overriding.

[IM04] The ENI Information Model should use the Single Responsibility Principle to design class hierarchies.

[IM05] The ENI Information Model should use the Liskov Substitution Principle to design class hierarchies.

[IM06] The ENI Information Model should use Design by Contract to manage behaviour.

[IM07] The ENI Information Model should use the Dependency Inversion Principle to ensure that the model is modular and extensible.

[IM08] The ENI Information Model should strive to realize loose coupling whenever possible to ensure that the model is modular and extensible.

[IM09] The ENI Information Model should strive to realize high cohesion whenever possible to ensure that the model is modular and extensible.

[IM10] The ENI Information Model should use the Role-Object Pattern to abstract individual entities into roles as applicable.

[IM11] The ENI Information Model shall have a single authoritative ENI information model.

NOTE: Requirement [IM11] means that at any time, only one ENI information model is authoritative. There may be other instances of the ENI information model, but all such instances represent changes to the authoritative ENI information model that have not yet been approved or incorporated into the authoritative ENI information model. These are called working information models.

[IM12] The authoritative ENI Information Model should have one or more backup instances.

[IM13] Each working ENI Information Model shall have one or more backup instances.

6.13 Data Model Requirements

[DM01] All ENI Data Models shall be derived from the ENI Extended Information Model.

[DM02] The ENI Information Model shall have a single authoritative ENI data model for each platform, language, and/or protocol.

NOTE 1: Since an ENI data model is specific to platform, language, and/or protocol, several ENI data models for a particular combination of platform, language, and/or protocol may exist. These are called working data models.

[DM03] There shall be zero or more instances of an ENI working data model present at any given time.

NOTE 2: Requirement [DM03] means that at any time, only one ENI data model is authoritative. There may be other instances of the ENI data model, but all such instances represent changes to the authoritative ENI data model that have not yet been approved or incorporated into the authoritative ENI data model. These are called working data models.

[DM04] The authoritative ENI data model should have one or more backup instances.

[DM05] Each working ENI data model shall have one or more backup instances.

[DM06] Generalization should be used to shorten class hierarchies.

NOTE 3: For example, all abstract super classes should be combined into a single class that contains their attributes, operations, and relationships.

[DM07] Specialization shall be used to refine class hierarchies.

NOTE 4: For example, a new subclass may be defined from an ENI authoritative data model in order to validate hypotheses (e.g. does this class simplify the resulting implementation) and/or optimise performance.

[DM08] If specialization is used in an ENI working data model, and those changes produce the desired results, then those changes shall be reflected back to the ENI authoritative data model in order to use them going forward.

[DM09] If changes are made to an ENI authoritative data model, then those changes shall be reflected back into an ENI working information model.

NOTE 5: Changes to an authoritative ENI data model may be made for two reasons:

1) to optimize that particular instance of an ENI data model; or

2) to improve the functionality of all ENI data models.

Changes of the first type need not be propagated back to a working ENI information model, as they are specific to that particular type of data model. In contrast, changes of the second type need to be propagated back to a working ENI information model because they can improve all derived ENI data models.

[DM10] Software patterns [i.5] should be used to construct an ENI Data Model.

6.14 Knowledge Graph Requirements

[KG1] A Knowledge Graph shall be grounded by a set of ontologies.

[KG2] A Knowledge Graph shall support deductive, inductive, and abductive inferencing.

[KG3] A Knowledge Graph should be used as a factual source of knowledge.

[KG4] Deductive reasoning shall be used to continuously validate an instance of a Knowledge Graph against its defined schema.

[KG5] A Knowledge Graph shall be used to integrate heterogeneous data sources.

[KG6] Semantic queries shall leverage the relationships and meanings encoded in a Knowledge Graph to understand user intent.

[KG7] Semantic queries of a Knowledge Graph shall be able to answer complex, multi-hop questions.

[KG8] A Knowledge Graph shall support context-sensitive queries.

[KG9] The reasoning capabilities in a Knowledge Graph should be used to help explain why decisions were made.

6.15 Transformer Architectural Requirements (normative)

6.15.1 Introduction

The following clauses define requirements for the ENI transformer architecture.

6.15.2 Transformer and LLM Usage in an ENI System

[TU1] Transformers and/or LLMs shall be used in an ENI System.

[TU2] Transformers and/or LLMs shall be used for constructing ENI Policies.

[TU3] Transformers and/or LLMs shall be used for translating natural language inputs into ENI Policies.

[TU4] Transformers and/or LLMs shall be used to provide services for a knowledge graph.

[TU5] A knowledge graph shall provide services for Transformers and/or LLMs.

[TU6] Transformers and/or LLMs shall be used in conjunction with a knowledge graph for providing solutions.

[TU7] Transformers based on T5 or FLAN-T5 shall be used to generate EBNF specifications for intent policies.

[TU8] BERT-based transformers shall be used to generate EBNF specifications for intent policies.

[TU9] Transformers based on CodeT5 or CodeT5+ shall be used to generate code from EBNF specifications for intent policies.

6.15.3 Transformer and LLM Architectural Requirements

[TA1] Transformers and/or LLMs shall be trained on as large a corpus of bimodal data as possible.

[TA2] Transformers and/or LLMs shall use any appropriate knowledge of the ENI System when creating, editing, deleting, or otherwise managing an ENI Policy.

[TA3] Transformers and/or LLMs shall use knowledge from ETSI GS ENI 019 (V4.1.1) [6] for constructing an ENI Policy.

[TA4] Transformers and/or LLMs shall not be directly exposed to systems external to ENI.

[TA4.1] Transformers and/or LLMs shall reside within a Functional Block in the ENI System Architecture.

[TA4.2] Transformers and/or LLMs shall use ENI Internal Reference Points to communicate with other ENI internal Functional Blocks.

[TA5] Transformers and/or LLMs shall use context information as defined in ETSI GS ENI 005 [2] for constructing an ENI Policy.

[TA5.1] The ENI System shall notify the Functional Block that contains Transformers and/or LLMs that it has updated context information.

[TA5.2] The ENI System shall make said context data available per the APIs of the selected Reference Point of the above communication.

[TA6] Transformers and/or LLMs shall use context information as defined in ETSI GS ENI 005 [2] for constructing an ENI Policy.

[TA6.1] The ENI System shall notify the Functional Block that contains Transformers and/or LLMs that it has updated situational awareness information.

[TA6.2] The ENI System shall make said situational awareness information available per the APIs of the selected Reference Point of the above communication.

[TA7] Transformers and/or LLMs shall use the grammar of a DSL when translating an ENI Policy to a DSL.

[TA8] Transformers and/or LLMs shall use RAG to increase the accuracy of generated text.

[TA9] Transformers and/or LLMs shall use prompt engineering to increase the accuracy of generated text.

6.15.4 Transformer and LLM Reference Point Requirements

[TR1] Transformers and/or LLMs shall use appropriate Internal Reference Points as defined in ETSI GS ENI 005 [2].

[TR2] Transformers and/or LLMs shall use appropriate External Reference Points as defined in ETSI GS ENI 005 [2].

6.16 Ontology Requirements

[ONT1] A set of ontologies shall be used to ground a Knowledge Graph.

[ONT2] A set of ontologies shall be used to provide a consensual lexicon.

[ONT3] The reasoning capabilities in a set of ontologies shall be used to help explain why decisions were made.

NOTE: Detailed ontology requirements are for further study.

6.17 Functional Requirements of Content Security Testing of LLM Against Prompt Injection Attacks

The following functional requirements shall be supported:

[CSTAPIA1] The ability to simulate diverse prompt injection attack scenarios.

[CSTAPIA2] The ability to generate representative and effective test datasets covering attack vectors and content security scenarios.

[CSTAPIA3] The ability to support API-based testing workflows for standardized interaction with LLMs under test.

[CSTAPIA4] The ability to verify LLM response mechanisms across different deployment environments (cloud/on-premises/edge).

[CSTAPIA5] The ability to integrate multi-level evaluation mechanisms for comprehensive security assessment.

[CSTAPIA6] The ability to measure evaluation metrics such as Attack Success Rate, Decline Rate, and Positive Rate with automated calculation capabilities.

6.18 Distributed Model Inference Service Scheduling Requirements

[DSSR.1] The ENI System shall support the scheduling and management of distributed AI/ML model inference services across heterogeneous compute resources including GPUs, CPUs, and AI acceleration cards.

NOTE: Heterogeneous compute resources may reside across edge nodes, regional data centres, and cloud infrastructure.

[DSSR.2] The ENI System shall use context information - including model architecture, parallelism strategy, current resource utilization, and SLA constraints - as inputs to inference scheduling decisions.

[DSSR.3] The ENI System architecture shall be flexible enough to support extensibility of scheduling strategies and parallelism modes without requiring redesign of core components.

[DSSR.4] The ENI System shall support the publication (registration) and discovery of distributed inference services, including their parallelism topology (TP degree, PP stages, EP routing) and resource requirements.

[DSSR.5] The ENI System shall support closed-loop control of inference service lifecycles, including deployment, scaling, health monitoring, and termination.

[DSSR.6] The ENI System shall invoke scheduling policies based on models that describe inference workload behaviour, including request arrival rates, token lengths, batch sizes, and SLA targets.

[DSSR.7] The ENI System shall support intent-based inference service publishing, allowing operators to express desired outcomes (e.g. target latency, minimum throughput) without specifying implementation details.

[DSSR.8] The ENI System shall recommend allocation or retrieval of compute resources (e.g. GPU nodes, memory, high-bandwidth interconnect bandwidth) for inference tasks in an intelligent manner to improve resource utilization efficiency.

[DSSR.9] The ENI System shall support dynamic computation of the resource configuration required to serve a given model with specified parallelism strategies (TP, PP, EP) and target performance metrics.

[DSSR.10] The ENI System shall be capable of performing proper planning of resource deployments to ensure that applicable scheduling policies and SLA constraints are not violated.

[DSSR.11] The ENI System shall use AI/ML-based prediction (e.g. time-series forecasting, reinforcement learning) to anticipate inference workload fluctuations and proactively adjust resource allocation.

[DSSR.12] The ENI System shall maintain a real-time view of available GPU resources, including per-node GPU memory availability, compute utilization, and interconnect bandwidth.

7 Non-functional requirements

7.1 Performance requirements

This clause captures requirements related to system performance, e.g. latency, accuracy, efficiency.

[PR.1] The ENI System shall be capable of analysing real-time data collected from a large number of devices, common in IoT environments, in order to ensure a proper performance according to SLA contracts.

7.2 Operational requirements

This clause captures operational requirements.

[OR.1] When using the ENI System to provide intelligent network operation and management, it shall be possible to minimize the Total Cost of Ownership (TCO), including Operating Expenses (OPEX) and Capital Expenses (CAPEX) of the network infrastructure. Existing network infrastructure shall be reused as much as possible to build or interwork with the ENI System.

[OR.2] The ENI System shall help to improve the energy efficiency of the whole network, where the information is available.

[OR.3] The ENI System shall provide, by using a functional block architecture, the support for multiple applications as used in different use cases.

7.3 Regulatory requirements

This clause captures regulatory requirements.

[RegR.1] The privacy of the users shall be properly protected during the collection, storage and analysis of the data.

NOTE: In Europe, the GDPR [i.3] applies.

[RegR.2] It shall be possible to analyse the network data without exposing its users.

[RegR.3] The provider of an ENI System shall comply with the provisions of the Lawful Interception regulations.

[RegR.4] The provider of an ENI System shall comply with the provisions of the data protection regulations.

[RegR.5] The provider of an ENI System shall comply with the provisions of the information security regulations.

7.4 Non-Functional Requirements of Content Security Testing of LLM Against Prompt Injection Attacks

The following non-functional requirements are applicable:

[CSTAPIANF1] Test Dataset Consistency: Datasets shall maintain version control and reproducibility.

[CSTAPIANF2] Test Dataset Currency: Datasets shall be updated on a regular basis to incorporate emerging attack vectors, trending real-world scenarios, and evolving adversarial techniques.

[CSTAPIANF3] Performance Scalability: Testing infrastructure shall handle concurrent test case execution with compliant response times.

[CSTAPIANF4] Security Isolation: Test environments shall implement measures to prevent cross-contamination with production systems.

[CSTAPIANF5] Traceability Compliance: Logging mechanisms shall capture all input data and LLM responses for forensic analysis.

[CSTAPIANF6] Adaptability to LLM Evolution: Testing frameworks shall allow easy update of attack patterns and evaluation criteria.

Annex A (informative): Change history

Date	Version	Information about changes
2017-05	v0.0.1	Initial early draft with skeleton
2017-06	v0.0.2	Combine ENI(17)000012r1
2017-07	v0.0.3	Combine ENI(17)000028r1
2017-08	v0.0.4	Combine ENI(17)000039r1, ENI(17)000044r1
2017-09	v0.0.5	Combine ENI(17)000079r1, ENI(17)000080r1
2017-10	v0.0.6	Combine ENI(17)003_029r1, ENI(17)003_031r1, ENI(17)003_032r1, ENI(17)003_035r1
2017-10	v0.0.7	Combine ENI(17)000086r1
2017-11	v0.0.8	Adding requirements from clause 4.12 of ETSI GR ENI 003
2017-11	v0.0.9	Change GR to GS, revise requirements in clauses 6 & 7
2017-12	v0.1.0	Combine ENI(17)004_007r1, ENI(17)004_020r1, ENI(17)004_021r1
2018-02	v0.1.1	Combine ENI(18)000042r1, ENI(18)000043r1
2018-03	v0.1.2	Combine ENI(18)005_019, ENI(18)005_020r1, ENI(18)005_022r3
2018-04	v1.1.1	Publication
2018-05	v2.0.0	Combine ENI(18)000105r1, ENI(18)000106r1
2018-06	v2.0.1	Combine ENI(18)006_024r2, ENI(18)006_027, ENI(18)006_026r4
2018-11	v2.0.2	Combine ENI(18)000186r1
2019-02	v2.0.3	Combine ENI(19)000021r1
2019-03	v2.0.4	Combine ENI(19)000046r1
2019-05	v2.0.5	Combine ENI(19)000121, ENI(19)000125r1
2019-07	V2.0.6	Combine ENI(19)000100r1 and changes agreed during ENI#10
2019-09	V2.1.1	Publication
2019-09	V3.0.1	Combine ENI(19)011_026r1 and new scope agreed during ENI#11
2019-12	V3.0.2	Combine ENI(19)012_028
2020-03	V3.0.3	Combine ENI(20)000041
2020-03	V3.0.4	Combine ENI(20)013_032r1
2020-04	V3.0.5	Combine ENI(20)000_008r1
2020-06	V3.0.6	Combine ENI(20)014_037r1
2020-12	V3.1.1	Publication
2021-11	V3.1.2	Combine ENI(21)000_214r1
2021-12	V3.1.3	Combine ENI(21)000_251
2022-03	V3.1.4	Combine ENI(22)021_009
2022-05	V3.1.5	Combine ENI(22)000_106r1
2022-06	V3.1.6	Combine ENI(22)000_133r1
2022-08	V3.1.7	Combine ENI(22)000_142
2022-09	V3.1.8	Combine agreed changes in ENI(22)023_029r3 and alignments
2023-09	V4.0.1	Initial draft for release 4.
2024-06	V4.0.2	Combine ENI(24)030_033
2024-11	V4.0.3	Combine ENI(24) 000_224
2025-06	V4.0.4	Combine ENI(25)034_042, ENI(25)034_043.
2025-11	V4.1.1	Combine ENI(25)000_139, ENI(25)000_233r1, ENI(25)000_237, ENI(25)000_239, ENI(25)000_240r1.

History

Version	Date	Status
V1.1.1	April 2018	Publication
V2.1.1	September 2019	Publication
V3.1.1	December 2020	Publication
V3.2.1	April 2023	Publication
V4.1.1	August 2026	Publication